Photutils Parameter Rationale

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The current spike-gated pipeline is deliberately conservative. Photutils finds compact bright residual-source candidates, but the bar-major intensity profile spike gate decides whether those candidates are allowed to affect the science profile.

The aim is to remove foreground-object contamination that causes narrow bar-profile spikes while preserving bar, ring, spiral-arm, nuclear, ansa, and shoulder structure. Earlier global Photutils runs showed that aggressive parameters can remove large amounts of valid galaxy light.

Current Parameter Interpretation

Parameter	Role and rationale
Masking model: spike-gated	Main protection layer. Photutils detections are candidates only; final masking requires intersection with detected bar-major profile spike samples.
Residual image	Photutils runs on science image minus a Gaussian-smoothed galaxy model, so compact sources stand out while broad galaxy light is suppressed.
Detection threshold	Brightness threshold in residual-sigma units. Lower values detect fainter objects but risk over-masking. Higher values are safer but can miss weaker foreground objects.
Smooth sigma: 15 px	Controls the broad galaxy model. Too small can absorb compact objects into the model or create galaxy-structure residuals; too large can leave broad gradients.
Connected-pixel minimum: 8 px	Rejects single-pixel noise and tiny artifacts. Lower values catch smaller sources but increase false positives.
Dilation radius: 3 px	Grows retained masks to include source wings. Larger values remove more foreground light but damage nearby galaxy data more quickly.
Max segment area: 500 px	Rejects large residual patches that are often galaxy arms, rings, or background structure rather than compact foreground objects.
Max elongation: 6	Rejects stretched segments, protecting against spiral arms, bars, streaks, and diffuse residuals.

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Central exclusion radius: 12 px	Protects the nucleus and nuclear rings, which are easy for residual-source detection to mistake for contaminants.
Profile width: 3 px	A narrow profile strip preserves local spikes while reducing single-pixel noise.
Bridge merge gap: 12 samples	Merges nearby masked profile gaps before straight log-linear interpolation. Too large can bridge valid data.

Optimisation Strategy

- Keep spike-gated mode as the safe science product for shoulder-recognition profiles.

- Optimise global Photutils masking separately, because it is trying to solve a broader image-cleaning problem.

- Use positive spike galaxies and no-spike control galaxies together; a parameter set must remove real spikes without changing clean profiles.

- Track masked segment count, masked pixel fraction, affected bar-profile samples, and profile-shape change.

- Tune detection_nsigma, dilation radius, max area, and smooth sigma first; npixels, max elongation, and central exclusion are secondary controls.

- Expect global Photutils masking to need stricter parameters than spike-gated masking, because it lacks the profile gate.

Practical Recommendation

For the production spike-gated run, 3.5 sigma can be acceptable because only segments that coincide with profile spikes are applied. For global Photutils runs, 3.5 sigma is expected to be aggressive and should be treated as a diagnostic comparison rather than a final science mask.

The proposed optimiser should score candidate parameter sets in tiers: first identify spike-positive performance, then penalise damage to no-spike controls, then inspect the best few parameter sets visually before processing the full sample.